

	FATE MONTHLY REPORT Doc.No : FATE-001 Version : 1.0 Date : 27-06-2024	
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FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
Annex
May 2024



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1 NIGHT1: Contingency tables

Tables in section 1 are included in the range: 1-22.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	3740	1373	99
	4.2 < ws < 7.6	2126	2654	1055
	ws > 7.6	224	1219	4957
Number of points 17447 ; PC=65.1%; EBD=1.9%; POD1=61.4%; POD2=50.6%; POD3=81.1%				

Table 1: night1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	5672	406	0
	4.2 < ws < 7.6	418	4556	253
	ws > 7.6	0	284	5858
Number of points 17447 ; PC=92.2%; EBD=0.0%; POD1=93.1%; POD2=86.8%; POD3=95.9%				

Table 2: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed ms^{-1}		Observations		
		ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	15304	159	0
	12.0 < ws < 18.0	90	1569	68
	ws > 18.0	0	26	231
Number of points 17447 ; PC=98.0%; EBD=0.0%; POD1=99.4%; POD2=89.5%; POD3=77.3%				

Table 3: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

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wind direction degree		North	East	South	Observations	West
Model data	North	6797	421	45		96
	East	526	831	438		5
	South	62	289	3189		266
	West	212	60	83		235
Number of points 13555 ; PC=81.5%; EBD=1.3%; POD1=89.5%; POD2=51.9%; POD3=84.9%; POD4=39.0%						

Table 4: night1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	7494	131	0		19
	East	93	1394	50		0
	South	0	76	3699		28
	West	10	0	6		555
Number of points 13555 ; PC=97.0%; EBD=0.0%; POD1=98.6%; POD2=87.1%; POD3=98.5%; POD4=92.2%						

Table 5: night1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	2182	728	29
	7.1 < rh < 15.3	2423	3940	949
	rh > 15.3	495	1712	4891
Number of points 17349 ; PC=63.5%; EBD=3.0%; POD1=42.8%; POD2=61.8%; POD3=83.3%				

Table 6: night1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	4664	249	3
	7.1 < rh < 15.3	435	5943	186
	rh > 15.3	1	188	5680
Number of points 17349 ; PC=93.9%; EBD=0.0%; POD1=91.5%; POD2=93.2%; POD3=96.8%				

Table 7: night1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations		
		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	16801	38	0
	50.0 < rh < 70.0	28	323	4
	rh > 70.0	0	7	148
Number of points 17349 ; PC=99.6%; EBD=0.0%; POD1=99.8%; POD2=87.8%; POD3=97.4%				

Table 8: night1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	2783	587
	1.9 < pwv < 3.5	647	3640
	pwv > 3.5	8	670
Number of points 16384 ; PC=82.6%; EBD=0.1%; POD1=80.9%; POD2=74.3%; POD3=88.4%			

Table 9: night1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	3362	113
	1.9 < pwv < 3.5	76	4713
	pwv > 3.5	0	71
Number of points 16384 ; PC=97.9%; EBD=0.0%; POD1=97.8%; POD2=96.2%; POD3=99.1%			

Table 10: night1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

FATE - Monthly Report

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2421	1428	453
	0.6 < see < 0.8	935	1201	1150
	see > 0.8	184	542	1180
Number of points 9494 ; PC=50.6%; EBD=6.7%; POD1=68.4%; POD2=37.9%; POD3=42.4%				

Table 11: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	3507	526	25
	0.6 < see < 0.8	399	2302	272
	see > 0.8	11	364	2560
Number of points 9494 ; PC=84.0%; EBD=0.4%; POD1=89.5%; POD2=72.1%; POD3=89.6%				

Table 12: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	3390	0	316
	0.6 < see < 0.8	0	3080	316
	see > 0.8	150	91	2467
Number of points 9494 ; PC=94.1%; EBD=4.9%; POD1=95.8%; POD2=97.1%; POD3=79.6%				

Table 13: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	3909	4	22
	0.6 < see < 0.8	0	3172	21
	see > 0.8	8	16	2835
Number of points 9494 ; PC=99.5%; EBD=0.3%; POD1=99.8%; POD2=99.4%; POD3=98.5%				

Table 14: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.24

FATE - Monthly Report

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1199	566	52
	$3.4 < \tau < 5.8$	495	1166	501
	$\tau > 5.8$	82	1063	3347
Number of points 8471 ; PC=67.4%; EBD=1.6%; POD1=67.5%; POD2=41.7%; POD3=85.8%				

Table 15: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1654	156	0
	$3.4 < \tau < 5.8$	122	2399	210
	$\tau > 5.8$	0	240	3690
Number of points 8471 ; PC=91.4%; EBD=0.0%; POD1=93.1%; POD2=85.8%; POD3=94.6%				

Table 16: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1563	41	216
	$3.4 < \tau < 5.8$	131	2177	216
	$\tau > 5.8$	82	577	3684
Number of points 8471 ; PC=87.6%; EBD=3.5%; POD1=88.0%; POD2=77.9%; POD3=89.5%				

Table 17: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1773	1	10
	$3.4 < \tau < 5.8$	3	2759	10
	$\tau > 5.8$	0	35	3890
Number of points 8471 ; PC=99.4%; EBD=0.1%; POD1=99.8%; POD2=98.7%; POD3=99.5%				

Table 18: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 1.22

FATE - Monthly Report

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1501	1392
	0.6 < glf < 0.7	797	1137
	glf > 0.7	273	386
Number of points 8624 ; PC=38.2%; EBD=16.8%; POD1=58.4%; POD2=39.0%; POD3=21.0%			

Table 19: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2250	420
	0.6 < glf < 0.7	312	2072
	glf > 0.7	9	423
Number of points 8624 ; PC=80.8%; EBD=0.4%; POD1=87.5%; POD2=71.1%; POD3=84.3%			

Table 20: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2268	380
	0.6 < glf < 0.7	30	2516
	glf > 0.7	273	19
Number of points 8624 ; PC=77.2%; EBD=17.9%; POD1=88.2%; POD2=86.3%; POD3=45.1%			

Table 21: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2556	19
	0.6 < glf < 0.7	6	2882
	glf > 0.7	9	14
Number of points 8624 ; PC=99.1%; EBD=0.5%; POD1=99.4%; POD2=98.9%; POD3=98.0%			

Table 22: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

2 DAY1: Contingency tables

Tables in section 2 are included in the range: 23-32.

wind speed ms^{-1}	Observations		
	ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	2334	430
	4.6 < ws < 7.4	2566	3439
	ws > 7.4	216	1992
Number of points 17043 ; PC=63.5%; EBD=1.7%; POD1=45.6%; POD2=58.7%; POD3=83.3%			

Table 23: day1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	4562	426
	4.6 < ws < 7.4	553	5055
	ws > 7.4	1	380
Number of points 17043 ; PC=89.9%; EBD=0.0%; POD1=89.2%; POD2=86.2%; POD3=94.2%			

Table 24: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	14605	166
	12.0 < ws < 18.0	131	1546
	ws > 18.0	0	64
Number of points 17043 ; PC=97.4%; EBD=0.0%; POD1=99.1%; POD2=87.0%; POD3=85.9%			

Table 25: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

FATE - Monthly Report

wind direction degree		North	East	South	Observations West
Model data	North	5178	360	16	99
	East	87	258	89	0
	South	12	45	2184	364
	West	2015	16	269	3912
Number of points 14904 ; PC=77.4%; EBD=0.3%; POD1=71.0%; POD2=38.0%; POD3=85.4%; POD4=89.4%					

Table 26: day1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations West
Model data	North	6957	57	0	205
	East	49	605	9	0
	South	0	17	2490	39
	West	286	0	59	4131
Number of points 14904 ; PC=95.2%; EBD=0.0%; POD1=95.4%; POD2=89.1%; POD3=97.3%; POD4=94.4%					

Table 27: day1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	1876	700	0
	8.9 < rh < 17.9	2251	4692	995
	rh > 17.9	231	1494	4782
Number of points 17021 ; PC=66.7%; EBD=1.4%; POD1=43.0%; POD2=68.1%; POD3=82.8%				

Table 28: day1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	4057	240	0
	8.9 < rh < 17.9	301	6464	159
	rh > 17.9	0	182	5618
Number of points 17021 ; PC=94.8%; EBD=0.0%; POD1=93.1%; POD2=93.9%; POD3=97.2%				

Table 29: day1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	16424	20	0
	50.0 < rh < 70.0	29	315	18
	rh > 70.0	0	9	206
Number of points 17021 ; PC=99.6%; EBD=0.0%; POD1=99.8%; POD2=91.6%; POD3=92.0%				

Table 30: day1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	2520	467	0
	2.1 < pwv < 3.7	840	3441	683
	pwv > 3.7	34	669	7690
Number of points 16344 ; PC=83.5%; EBD=0.2%; POD1=74.2%; POD2=75.2%; POD3=91.8%				

Table 31: day1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	3273	96	0
	2.1 < pwv < 3.7	121	4414	87
	pwv > 3.7	0	67	8286
Number of points 16344 ; PC=97.7%; EBD=0.0%; POD1=96.4%; POD2=96.4%; POD3=99.0%				

Table 32: day1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

3 NIGHT2: Contingency tables

Tables in section 3 are included in the range: 33-42.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	3531	1311	93
	4.2 < ws < 7.6	2233	2651	998
	ws > 7.6	252	1221	4782
Number of points 17072 ; PC=64.2%; EBD=2.0%; POD1=58.7%; POD2=51.1%; POD3=81.4%				

Table 33: night2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	6647	480	66
	East	547	777	632
	South	41	268	2898
	West	103	79	82

Number of points 13177 ; PC=79.6%; EBD=1.5%; POD1=90.6%; POD2=48.4%; POD3=78.8%; POD4=29.6%

Table 34: night2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	1888	696
	7.1 < rh < 15.3	2482	3533
	rh > 15.3	642	1948

Number of points 16950 ; PC=59.5%; EBD=4.1%; POD1=37.7%; POD2=57.2%; POD3=80.9%

Table 35: night2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	2805	715
	1.9 < pwv < 3.5	669	3133
	pwv > 3.5	19	883

Number of points 16168 ; PC=80.1%; EBD=0.4%; POD1=80.3%; POD2=66.2%; POD3=88.3%

Table 36: night2 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2662	1472	581
	0.6 < see < 0.8	1118	1410	1232
	see > 0.8	290	855	1603
Number of points 11223 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 37: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	3855	0	397
	0.6 < see < 0.8	0	3557	397
	see > 0.8	215	180	3019
Number of points 11223 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 38: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1525	581	61
	$3.4 < \tau < 5.8$	578	1344	528
	$\tau > 5.8$	97	1325	4054
Number of points 10093 ; PC=68.6%; EBD=1.6%; POD1=69.3%; POD2=41.4%; POD3=87.3%				

Table 39: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1939	49	268
	$3.4 < \tau < 5.8$	164	2282	268
	$\tau > 5.8$	97	919	4375
Number of points 10093 ; PC=85.2%; EBD=3.6%; POD1=88.1%; POD2=70.2%; POD3=89.1%				

Table 40: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1557	1643
	0.6 < glf < 0.7	1063	1518
	glf > 0.7	250	375
Number of points 10289 ; PC=38.1%; EBD=15.6%; POD1=54.3%; POD2=42.9%; POD3=21.7%			

Table 41: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2585	384
	0.6 < glf < 0.7	35	3128
	glf > 0.7	250	24
Number of points 10289 ; PC=78.8%; EBD=16.9%; POD1=90.1%; POD2=88.5%; POD3=46.9%			

Table 42: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

4 DAY2: Contingency tables

Tables in section 4 are included in the range: 43-46.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	2427	438	101
	4.6 < ws < 7.4	2547	3452	1140
	ws > 7.4	207	1932	4936
Number of points 17180 ; PC=63.0%; EBD=1.8%; POD1=46.8%; POD2=59.3%; POD3=79.9%				

Table 43: day2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	5074	368	19
	East	152	210	86
	South	16	36	2012
	West	2094	40	478

Number of points 14984 ; PC=74.6%; EBD=0.5%; POD1=69.2%; POD2=32.1%; POD3=77.5%; POD4=88.2%

Table 44: day2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	1726	479
	8.9 < rh < 17.9	2596	4258
	rh > 17.9	277	2072

Number of points 17155 ; PC=62.6%; EBD=1.7%; POD1=37.5%; POD2=62.5%; POD3=82.7%

Table 45: day2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	2558	566
	2.1 < pwv < 3.7	1200	3051
	pwv > 3.7	41	864

Number of points 16510 ; PC=78.3%; EBD=0.3%; POD1=67.3%; POD2=68.1%; POD3=88.8%

Table 46: day2 - Contingency table for precipitable water vapor (mm) in standard configuration

5 NIGHT3: Contingency tables

Tables in section 5 are included in the range: 47-56.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	3483	1411	158
	4.2 < ws < 7.6	2232	2540	1123
	ws > 7.6	327	1174	4557
Number of points 17005 ; PC=62.2%; EBD=2.9%; POD1=57.6%; POD2=49.6%; POD3=78.1%				

Table 47: night3 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	6484	562	81
	East	548	745	640
	South	57	247	2891
	West	115	50	104

Number of points 13081 ; PC=78.7%; EBD=1.5%; POD1=90.0%; POD2=46.4%; POD3=77.8%; POD4=32.1%

Table 48: night3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	1689	786
	7.1 < rh < 15.3	2577	3502
	rh > 15.3	712	1915

Number of points 16883 ; PC=56.7%; EBD=4.5%; POD1=33.9%; POD2=56.5%; POD3=76.9%

Table 49: night3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	2765	900
	1.9 < pwv < 3.5	673	2794
	pwv > 3.5	57	1034

Number of points 16101 ; PC=76.0%; EBD=0.9%; POD1=79.1%; POD2=59.1%; POD3=84.8%

Table 50: night3 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2649	1562	576
	0.6 < see < 0.8	1181	1333	1291
	see > 0.8	243	832	1509
Number of points 11176 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 51: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	3868	0	407
	0.6 < see < 0.8	0	3596	407
	see > 0.8	205	131	2969
Number of points 11176 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 52: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1531	582	44
	$3.4 < \tau < 5.8$	567	1287	603
	$\tau > 5.8$	103	1378	4000
Number of points 10095 ; PC=67.5%; EBD=1.5%; POD1=69.6%; POD2=39.6%; POD3=86.1%				

Table 53: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1916	45	283
	$3.4 < \tau < 5.8$	182	2327	283
	$\tau > 5.8$	103	875	4364
Number of points 10095 ; PC=85.3%; EBD=3.8%; POD1=87.1%; POD2=71.7%; POD3=88.5%				

Table 54: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1476	1522
	0.6 < glf < 0.7	1149	1623
	glf > 0.7	251	384
Number of points 10291 ; PC=37.6%; EBD=16.2%; POD1=51.3%; POD2=46.0%; POD3=19.8%			

Table 55: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2571	328
	0.6 < glf < 0.7	54	3176
	glf > 0.7	251	25
Number of points 10291 ; PC=78.8%; EBD=17.3%; POD1=89.4%; POD2=90.0%; POD3=45.5%			

Table 56: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

6 DAY3: Contingency tables

Tables in section 6 are included in the range: 57-60.

wind speed <i>ms⁻¹</i>		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	2387	565	122
	4.6 < ws < 7.4	2481	3410	1157
	ws > 7.4	264	1875	4852
Number of points 17113 ; PC=62.2%; EBD=2.3%; POD1=46.5%; POD2=58.3%; POD3=79.1%				

Table 57: day3 - Contingency table for wind speed (*ms⁻¹*) in standard configuration

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wind direction degree	Observations			
	North	East	South	West
Model data	North	4922	411	41
	East	126	178	95
	South	13	30	1963
	West	2269	41	496

Number of points 14939 ; PC=72.2%; EBD=0.6%; POD1=67.1%; POD2=27.0%; POD3=75.6%; POD4=85.6%

Table 58: day3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	1459	586
	8.9 < rh < 17.9	2830	4025
	rh > 17.9	290	2181

Number of points 17088 ; PC=58.3%; EBD=1.9%; POD1=31.9%; POD2=59.3%; POD3=78.5%

Table 59: day3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	2385	667
	2.1 < pwv < 3.7	1348	2881
	pwv > 3.7	67	847

Number of points 16490 ; PC=74.8%; EBD=1.2%; POD1=62.8%; POD2=65.6%; POD3=85.2%

Table 60: day3 - Contingency table for precipitable water vapor (mm) in standard configuration